

ELEC 401  
Analog CMOS Integrated Circuit Design  
Design Project: Two-Stage Opamp Design

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# Chapter 1

## Introduction

Operational Amplifiers (op-amps) are fundamental active devices used by analog designers to amplify an input voltage signal fed in by two input terminals and produce a single output voltage. The op-amp is usually characterized by its open loop gain, bandwidth, stability and power consumption. Op-amps usually have multiple stages to achieve higher voltage gain, sufficient output swing and reduce noise. An ideal Op-Amp has infinite open-loop gain, infinite input resistance, zero output resistance, infinite bandwidth and zero offset. Op-amps are often used for signal conditioning, amplification, filtering and performing mathematical operations in analog circuits.

This project presents the design and simulation of a two-stage differential to single-ended op-amp based on the given topology in figure 1 in the 45-nm CMOS process. Additionally, the amplifier employs a common mode feedback(CMFB) circuit to regulate the output common-mode voltage. Miller compensation with a series resistor ( $R_C$ ) was used for tweaking the phase margin(PM) and unity gain frequency(UGF) for stability.

This report will begin with a description of the design objectives, followed by the design process using Cadence EDA tools and lastly the overall performance achieved of the op-amp will be evaluated against key performance metrics.

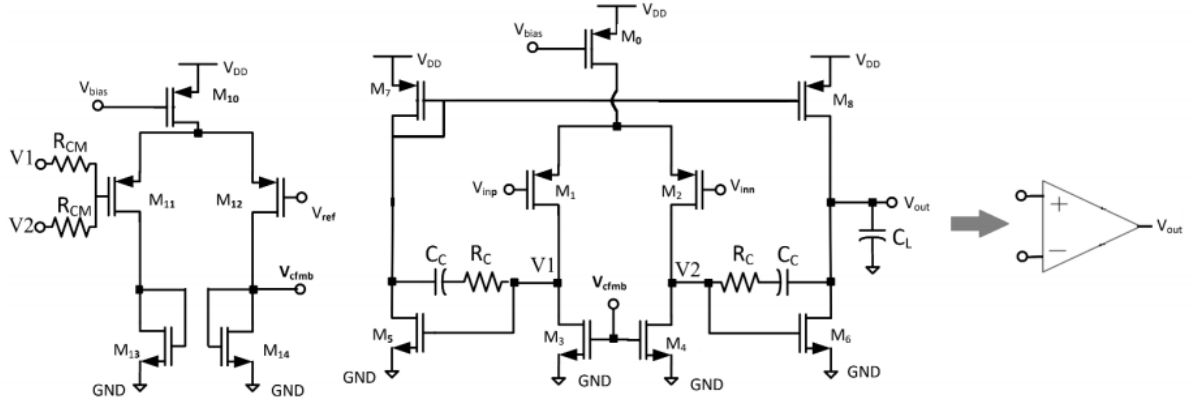


Figure 1.1: Provided op-amp topology

*NOTE:  $V_{inp}$  &  $V_{inn}$  inputs were found to be switched.*

# Chapter 2

## Design Procedure

### 2.1 Methodology

The project was a significant learning process on how to design analog circuits. Many mistakes were made but provided key learning opportunities that will be carried forward to future projects.

Firstly, the power budget had to be considered. The CMFB circuit's requirement was simply  $\leq 40 \mu W$ . Therefore, we can calculate the maximum current that can be provided to the CMFB circuit.

$$I_{cmfb} \leq \frac{40 \mu W}{1.0 V}$$
$$I_{cmfb} \leq 40 \mu A$$

Considering the two-stage amplifier circuit where  $I_1$  is the current through stage 1 and  $I_2$  is the current through stage 2:

$$I_{max} = I_1 + I_2$$
$$I_{max} = \frac{40 mW}{1.0 V} = 400 \mu A$$

We have a maximum of  $400 \mu A$  that can be distributed to both stages 1 and 2. Initially, it was thought that stage 1 should be provided with more current than stage 2, however it was found later into the design process that this made it significantly harder to bias the input stage transistors  $M_1$  and  $M_2$  as well as  $M_{11}$  in the CFMB circuit into saturation together. This pushed the design to have more current in stage 2 which would also make sense as we would like to have a high enough current to boost the stage 2 current source amplifier's transconductance, specifically of  $M_6$ .

This brings us to ensuring the slew rate has been met as it is defined as the maximum rate of output change when the input signal is large.

$$SR = \frac{I_1}{C_c}$$

Initially,  $C_c$  was set to  $1 pF$ , thus for a minimum slew rate of  $20 V/\mu s$  it must be ensured that  $I_1$  is larger than  $20 \mu A$ .

Moving onto the small signal differential gain, we can examine the small signal circuits of stage 1 and stage 2 as seen in figure 2.1 below to obtain an expression in terms of transconductance and output resistances of the transistors.



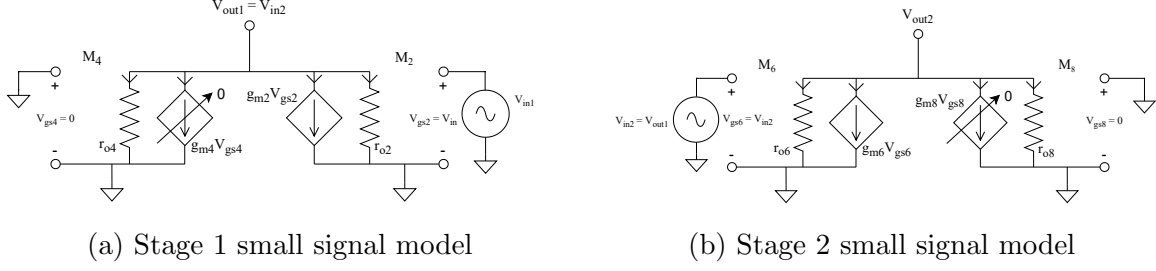


Figure 2.1: Small signal models for the two amplifier stages

Analyzing stage 1:

$$\begin{aligned} \frac{V_{out1}}{r_{o4}} + \frac{V_{out1}}{r_{o2}} + g_{m2}V_{gs2} &= 0 \\ V_{out1} \left( \frac{1}{r_{o4}} + \frac{1}{r_{o2}} \right) &= -g_{m2}V_{in1} \\ \frac{V_{out1}}{V_{in1}} &= -g_{m2} (r_{o4} \parallel r_{o2}) \end{aligned}$$

Analyzing stage 2:

$$\begin{aligned} \frac{V_{out2}}{r_{o6}} + \frac{V_{out2}}{r_{o8}} + g_{m6}V_{gs6} &= 0 \\ V_{out2} \left( \frac{1}{r_{o6}} + \frac{1}{r_{o8}} \right) &= -g_{m6}V_{in2} \\ \frac{V_{out2}}{V_{in2}} &= -g_{m6} (r_{o6} \parallel r_{o8}) \end{aligned}$$

Calculating the overall amplifier small-signal differential gain:

$$\begin{aligned} A_T &= A_1 \times A_2 \\ &= -g_{m2} (r_{o4} \parallel r_{o2}) \times -g_{m6} (r_{o6} \parallel r_{o8}) \end{aligned} \quad (2.1)$$

This leads us to equation 2.1 where the overall gain is dependent on the transconductances of transistors M<sub>2</sub> and M<sub>6</sub> as well as the output resistances of M<sub>2</sub>, M<sub>4</sub>, M<sub>6</sub> and M<sub>8</sub>. While simulating in Cadence, it was evident that a larger drain current causes a higher transconductance, but a lower output resistance. This makes sense as we can recall from the small signal model equations that:

$$r_o = \frac{1}{\lambda I_D} \quad g_m = \sqrt{2\mu C_{ox} \frac{W}{L} I_D} \quad I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

Before optimizing the gain of the amplifier, we must first bias stages 1 and 2 by selecting  $V_{bias}$  and  $V_{ref}$  necessary for all transistors to operate in the saturation region. This was done by making use of Cadence's Spectre's simulation environment to sweep variables, plot graphs, and check regions of operation of the transistors while satisfying the necessary current requirements for each stage. It was highly important to use the necessary equations required for biasing nmos and pmos devices into the saturation region:

$$\begin{aligned} \text{NMOS: } V_{gs} &> V_{th}, & V_{ds} &\geq V_{gs} - V_{th} \\ \text{PMOS: } V_{sg} &> |V_{th}|, & V_{sd} &\geq V_{sg} - |V_{th}| \end{aligned}$$

For a diode connected mos:  $V_{gs} \geq V_{th}$  leads to saturation as  $V_{gs} = V_{ds}$ .

Firstly, stage 1 was loaded onto Cadence and simulations were run to obtain the threshold voltage of the pmos1v\_LVT and nmos1v\_LVT transistors. After obtaining  $V_{th,n}$  and  $V_{th,p}$  pertaining to the nmos and pmos respectively, the circuit was initially designed with a  $V_{ref}$  and  $V_{bias}$  of 0.5V. Aspect ratios were chosen arbitrarily for the purposes of getting familiar with the circuit.

After stage 1 was biased roughly, stage 2 was added, followed by the CMFB circuit. Following the initial simulations, the aspect ratio of the transistors were tweaked to control the drain currents and come to finalize  $V_{\text{bias}}$  of 0.46V and  $V_{\text{ref}}$  of 0.3V.

Oftentimes, a common issue was not having a high enough drain-source voltage of the stage 1 nmos pair ( $V_{\text{ds4}}$ ) for biasing  $M_6$  into saturation. This was fixed by decreasing the aspect ratio ( $W/L$ ) to lower the drain current through  $M_4$  which led to a higher  $V_{\text{ds4}}$ .

Coming back to the stage gain, making use of the simulation environment was highly advantageous as it was possible to sweep variables like  $W$  and  $L$  to determine the optimal gain while within the maximum current allowed for the respective stage. Specific widths and lengths of the transistors were chosen so that there was a good balance between a small enough  $g_m$  that would allow us to have as large of an output resistance that we can for each transistor. Ignoring  $g_{m4}$  and  $g_{m8}$  were possible for the purposes of the small-signal gain, however making sure that they were still in saturation.

After optimising the gain within power requirements, the circuit in figure 2.2 was designed. Figure 2.3 shows the schematic on cadence with the dc-op values annotated. Figure 2.3 displays the transconductances and output resistances of each transistor, allowing us to calculate the theoretical small-signal differential gain using the expression derived before, showing that the design requirement for gain has been met:

$$\begin{aligned} A_T &= -1.06036 \times 10^{-3} (28.5872k \parallel 31.3777k) \times -1.59238 \times 10^{-3} (28.5872k \parallel 31.3777k) \\ &= 377.8255 \text{ V/V} = 51.55 \text{ dB} \end{aligned}$$

At this point, the total current consumed by the CMFB circuit is 40.3535  $\mu\text{A}$  while the amplifier consumes a total of 375.293  $\mu\text{A}$ . Both values fall within the given design constraints.

Now, examining the open-loop transfer function will allow us to ensure low-frequency gain, PM, and UGF targets are met. With  $R_c = 0 \Omega$  and  $C_c = 1 \text{ pF}$ , the UGF and PM were too low at approximately 155 MHz, and 11° respectively. Keeping the miller capacitor at 1pF and increasing  $R_c$  to 1M lead to an increase in the UGF to 955 MHz, however PM remained relatively unchanged. At this point, while the UGF requirement is met, the PM will still need to increase. Instead, if we kept  $R_c$  at 0 and increased  $C_c$  from 1pF to 10pF, it was evident that this decreases the UGF, to 35.68 Mhz, but increases the PM to 27°. Following these observations, it would also be useful to test the effect of changing the width of the stage 2 mos devices as the zero we are placing depends on the transconductance of the stage 2 mos devices. We can test this by sweeping both the multipliers of the top pmos and bottom nmos in stage 2. This yielded figures 2.2 and 2.3 where it was realised that as we increase the width of the bottom nmos device, this increases the PM. However increasing the top pmos device provided minimal change in UGF or PM.

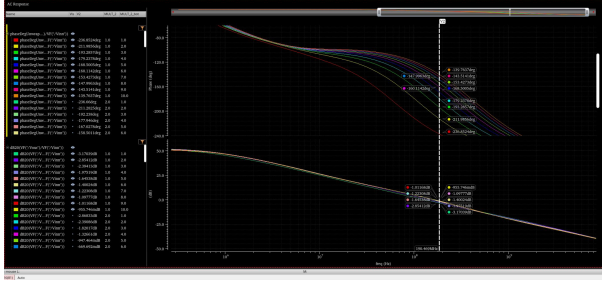


Figure 2.2: Effect of increasing the stage 2 nmos multiplier on PM and UGF

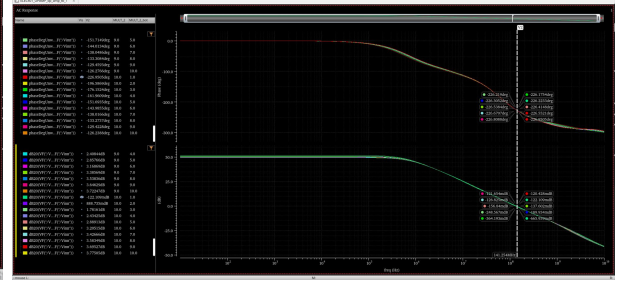


Figure 2.3: Effect of increasing the stage 2 pmos multiplier on PM and UGF

In order to minimise changes to the circuit, it was first attempted to achieve the PM and UGF using solely the compensation capacitor and resistor. Thus sweeping the capacitor and resistor lended the following graph that was used to recognise that a smaller resistor seemed better to improve the PM while maintaining a high enough UGF to meet the design requirement. A small capacitor also helps with having minimal decrease in the bandwidth as seen in figure 2.5.



Figure 2.4: Effect of varying  $R_c$  and  $C_c$

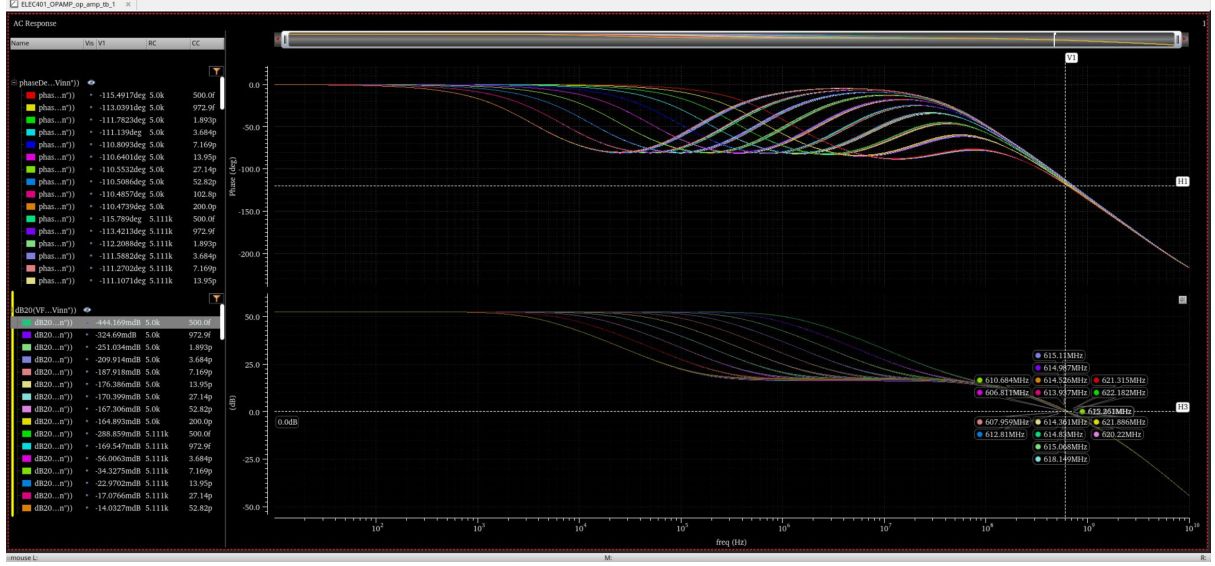


Figure 2.5: Effect of higher  $C_c$  on reducing bandwidth while  $R_c$  kept constant

Finally, running further simulations, to narrow down values, a compensation resistor and capacitor value of 5.44k and 1.90p were selected to nail the design requirements down to a UGF of 618 MHz and PM of 65.38°(refer to section 3.2). Now, calculating the slew rate to ensure it is greater than 20 V/μs:

$$\begin{aligned} SR &= \frac{92.055\mu A}{1.90pF} \\ &= 48.45V/\mu s \end{aligned}$$

Something that should have been considered earlier in the design process is the differential output swing of the amplifier as meeting this target requires resizing the second stage. To find the upper and lower bounds of the output voltage:

Consider M8:

$$\begin{aligned} V_{sd8} &\geq V_{sg8} - |V_{th8}| \\ V_{d8} &\leq V_{g8} + |V_{th8}| \\ V_{d8} &\leq 445.986m + |-324.189m| \\ V_{d8} &\leq 770.175m \end{aligned}$$

Consider M6:

$$\begin{aligned} V_{ds6} &\geq V_{gs6} - |V_{th6}| \\ V_{d8} &\geq 491.633m - |368.171m| \\ V_{d8} &\geq 123.462m \end{aligned}$$

This effectively lends an output voltage swing of 647mV. In order to meet the requirement of an output swing  $\geq 0.75$ , it would be possible to increase the upper bound by increasing the gate voltage of the pmos in the second stage. This would require increasing the drain-source voltage of the second stage nmos by decreasing W/L to drop the current through stage 2. This would sacrifice some gain due to a decrease in transconductance, however as the output resistance increases, this loss in gain should be minimal.

## 2.2 Design Schematic

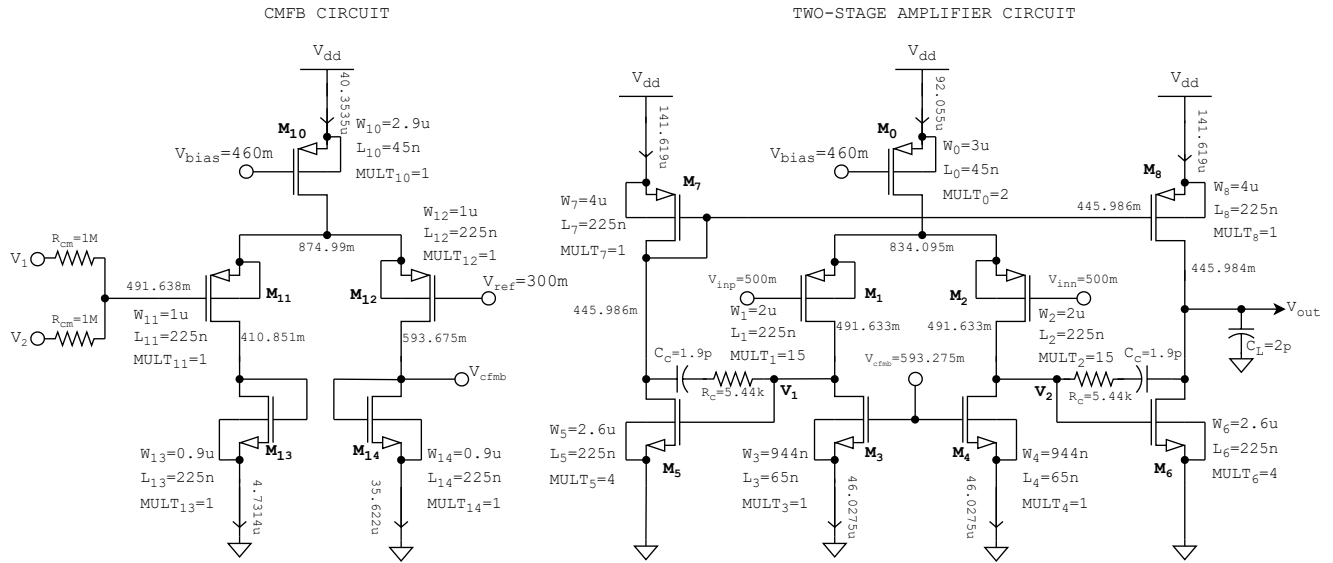


Figure 2.6: Diagram of component parameters

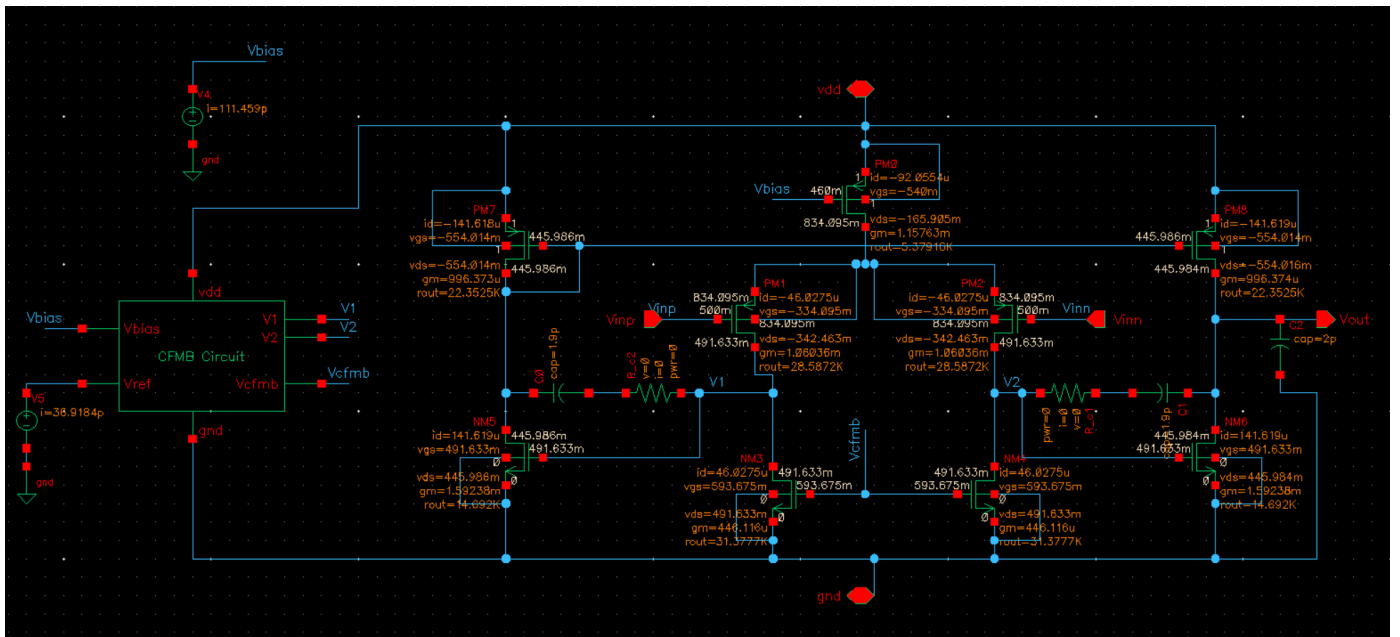


Figure 2.7: Amplifier circuit dc-op with MOS output resistance annotated

## 2.3 Overall Performance

All design requirements other than differential output swing have been met as noted below.

| Specification                                    | Requirement                       | Achieved         |
|--------------------------------------------------|-----------------------------------|------------------|
| $V_{DD}$                                         | 1.0 V                             | 1.0 V            |
| GND                                              | 0 V                               | 0 V              |
| $C_L$                                            | 2 pF                              | 2 pF             |
| Nominal input common-mode (input DC level)       | $V_{DD}/2=0.5$ V                  | 0.5 V            |
| Nominal output common-mode (output DC level)     | $V_{DD}/2=0.5$ V                  | 0.446 V          |
| Two-stage power consumption                      | $\leq 0.4$ mW                     | 0.375 mW         |
| CMFB circuit power                               | $\leq 40$ $\mu$ W                 | 40.35 $\mu$ W    |
| Differential output Swing                        | $\geq 0.75$ V                     | 0.65 V           |
| Low-frequency differential gain                  | $\geq 46$ dB                      | 52.44 dB         |
| Small-signal unity gain frequency                | $\geq 600$ MHz                    | 618 MHz          |
| Phase Margin                                     | $\geq 60^\circ$ & $\leq 90^\circ$ | 65.39°           |
| Slew Rate                                        | $\geq 20$ V/ $\mu$ s              | 48.45 V/ $\mu$ s |
| Maximum length of transistors ( $L_{min}=45$ nm) | $5 \times L_{min}=225$ nm         | 225 nm           |
| Maximum width per Multiplier                     | 4 $\mu$ m                         | 4 $\mu$ m        |
| Maximum Multiplier of transistors                | 50                                | 15               |

## 2.4 CMFB Architecture

The CMFB circuit provides a feedback loop that controls and stabilises the output common-mode(CM) voltage of the amplifier. If the gate of  $M_{11}$  is equal to the reference voltage, current is divided equally through the CMFB circuit. However, if the output CM is too high (greater than  $V_{ref}$ ), the CMFB circuit will pull it back down as the source to gate voltage of  $M_{11}$  drops, thereby decreasing the current through the branch and dropping the  $V_{cfmb}$ . This drop in  $V_{cfmb}$  will decrease the current going through stage 1, and so increasing the CM differential output voltage of the amplifier. The opposite happens when the output CM is lower than  $V_{ref}$ .  $V_{ref}$  is not necessarily the output common mode voltage we desire (0.5V), rather it is an internal reference for the CMFB circuit due to our design choices.

# Chapter 3

## Performance Results

### 3.1 DC-OP Point

By feeding in a 0.5V common-mode input, a dc-op simulation on Cadence yielded the following results where region 2 represents saturation.

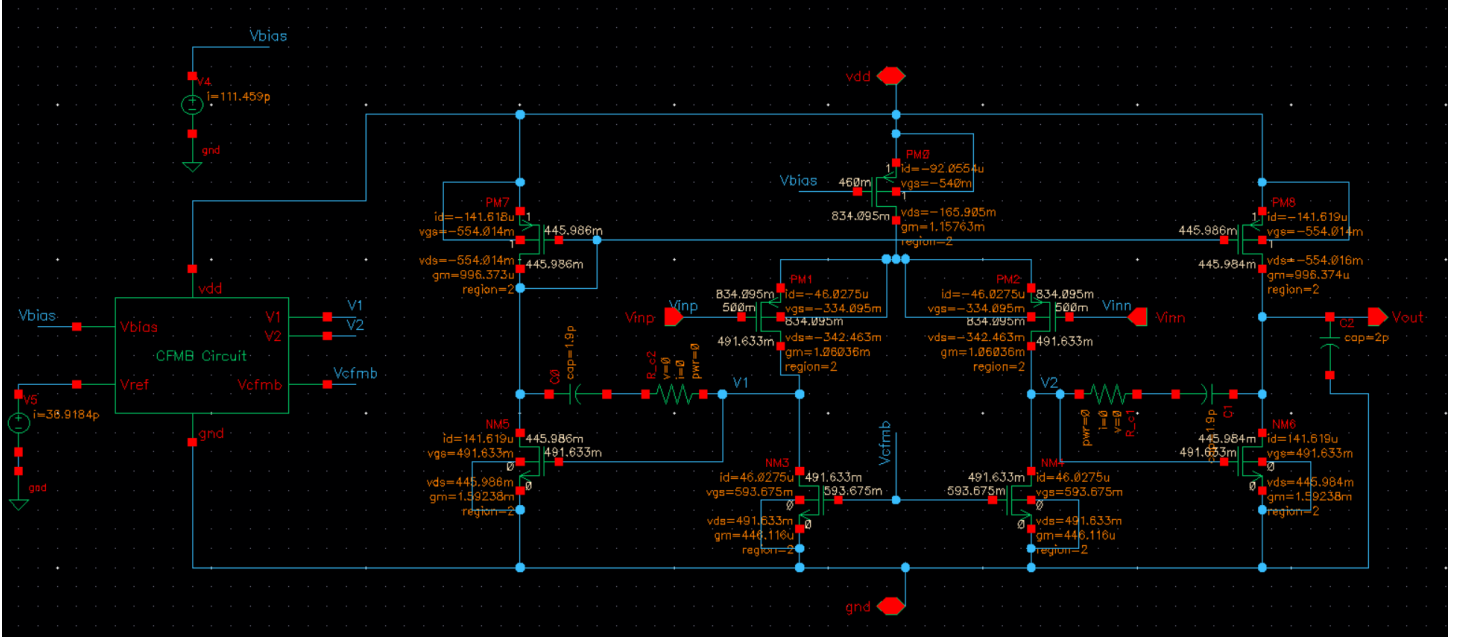


Figure 3.1: Amplifier Circuit DC-OP With Regions Annotated

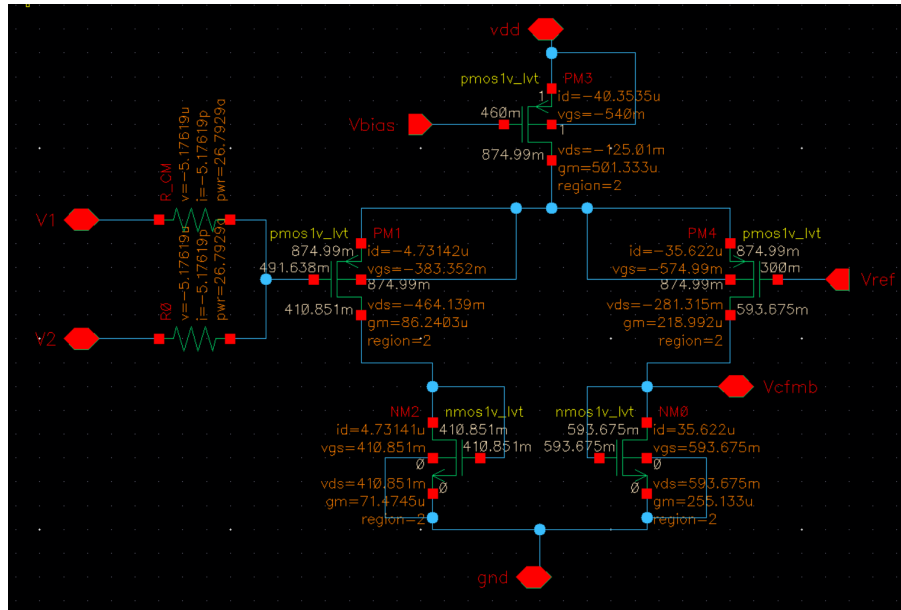


Figure 3.2: CMFB Circuit DC-OP With Regions Annotated



## 3.2 Open-Loop Transfer Function

The following bode plot was obtained with a differential small-signal AC input:

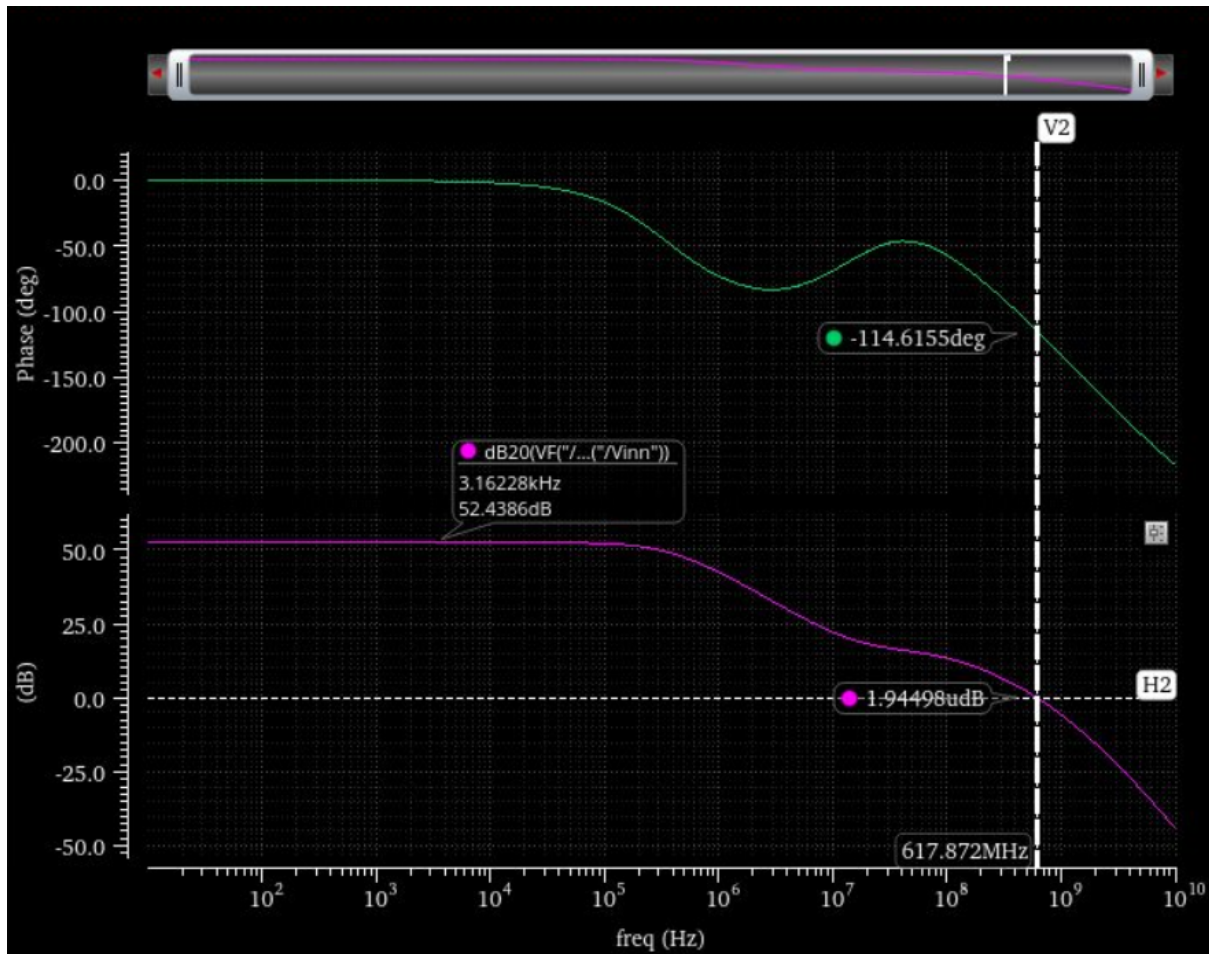


Figure 3.3: Open-loop transfer function

### 3.2.1 Phase Margin

The phase margin is the difference between the phase at UGF and  $-180^\circ$ . As seen in figure 3.3:

$$PM = 180^\circ - 114.6155^\circ = 65.3845^\circ$$

This meets our design requirement of  $\geq 60^\circ$  &  $\leq 90^\circ$ .

The plot also shows that a UGF of 617.872 MHz was achieved, which meets the design requirement of  $\geq 600$  MHz.

### 3.2.2 Low-Frequency Gain

As annotated in figure 3.3, the low frequency gain is approximately 52.44 dB.



### 3.3 Closed-Loop Transfer Function

The following circuit in figure 3.3 was used to create a bode plot of the closed-loop system and find the 3-dB frequency. Vin1 is an AC source with an Amplitude of 1u. Figure 3.4 is produced which presents the closed-loop AC gain of 10.65 mdB. The resulting 3-dB frequency of the system is 390.90 MHz where -2.989 dB lies.

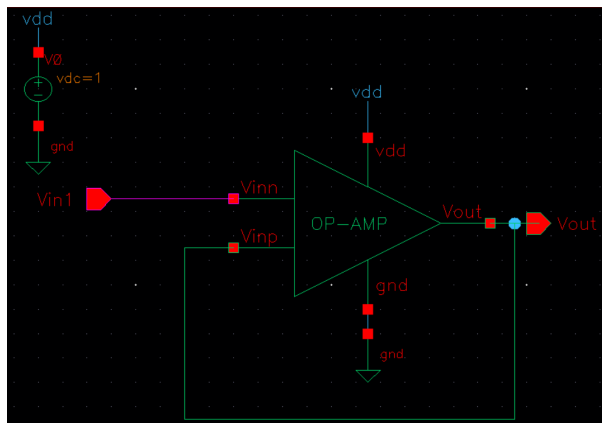


Figure 3.4: Designed op-amp in unity buffer configuration

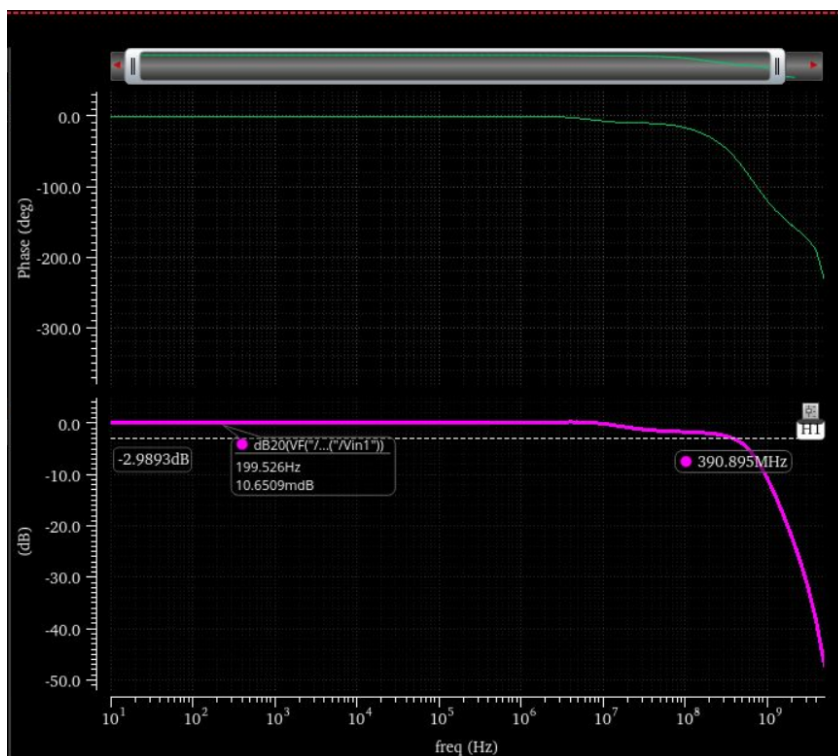
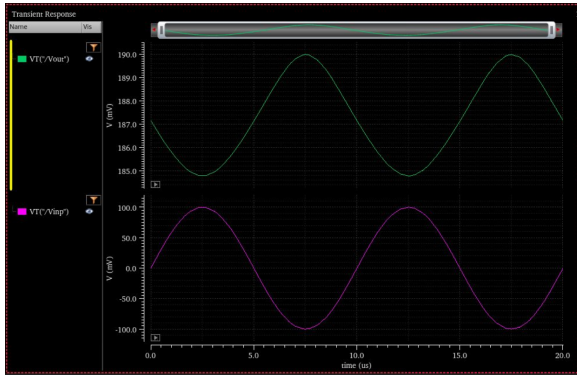


Figure 3.5: Closed-loop Transfer Function with 3-dB frequency annotated

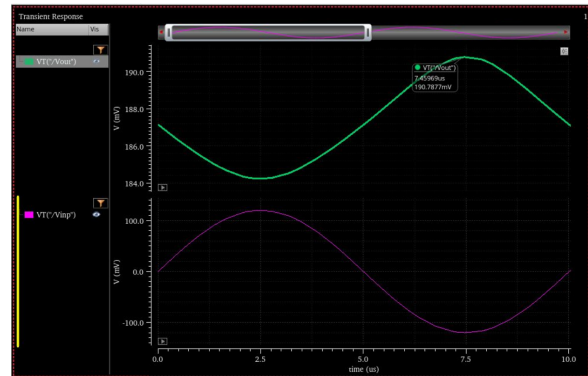
### 3.4 Maximum Undistorted Output Swing

Firstly, a 0.1 MHz differential sinusoidal input is applied to the open-loop circuit with an appropriately small amplitude that avoids distortion at the input. We then gradually increase the amplitude until distortion occurs. Making use of the simulation environment, an amplitude sweep was run to look at the approximate voltage where distortion occurs. As seen in figure 3.5(d), distortion starts to appear as the peak of the wave-

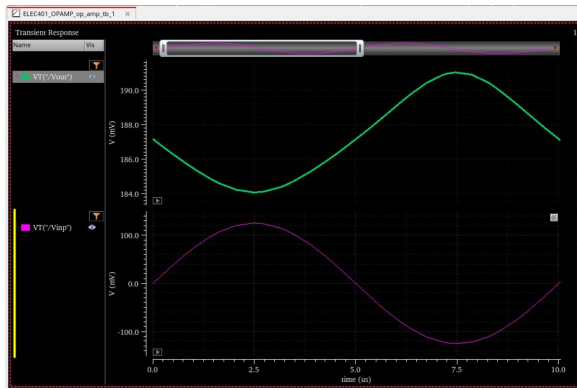
form gets chopped off. Thus, the maximum undistorted output swing was found to be approximately 125mV.



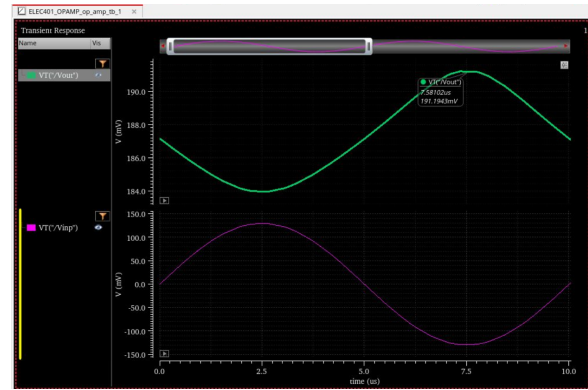
(a) 100mV



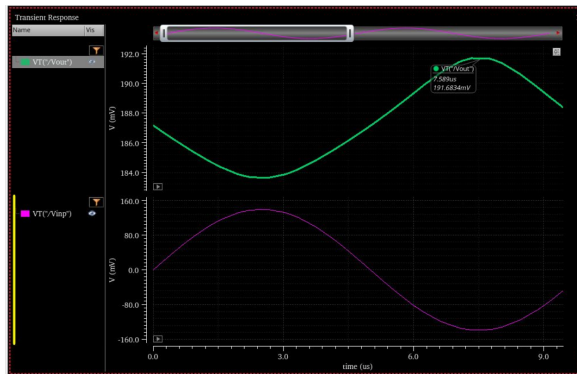
(b) 120mV



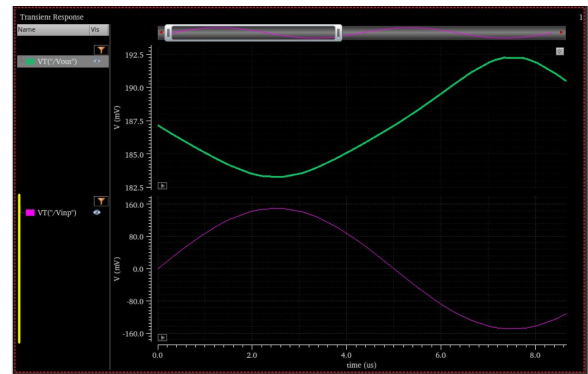
(c) 125mV



(d) 130mV



(e) 140mV



(f) 150mV

Figure 3.6: Transient waveforms of 0.1 MHz differential sinusoidal input

### 3.5 Transient Responses: Settling Behaviour & Slew rate

To plot the transient response of the step functions, the op-amp is connected as in figure 3.5. In order to calculate the small-signal 10 to 90% settling time, we first find the difference between the final settling value of the output voltage waveform and the input voltage waveform. After finding the difference, we add  $0.9 \times \Delta V$  or  $0.1 \times \Delta V$  to get the 90% or 10% voltage values respectively. The large-signal initial rising/falling slopes can simply be calculated by the rise/run of the initial linear region.

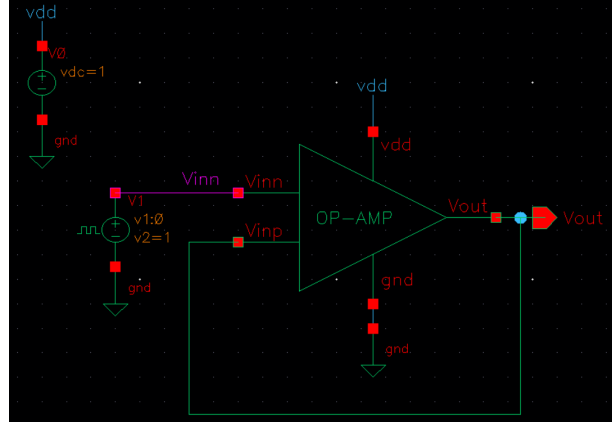


Figure 3.7: Step-response circuit configuration

#### 3.5.1 0.495V - 0.505V

Calculating the 10 to 90% settling time as seen in figure 3.7:

$$106.0863\text{ns} - 100.4288\text{ns} = 5.6575\text{ns}$$

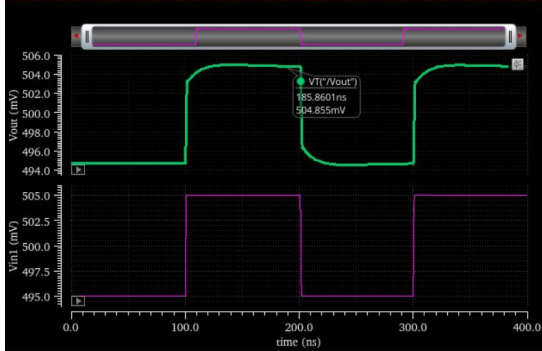


Figure 3.8: Closed-loop step response (495mV–505mV)

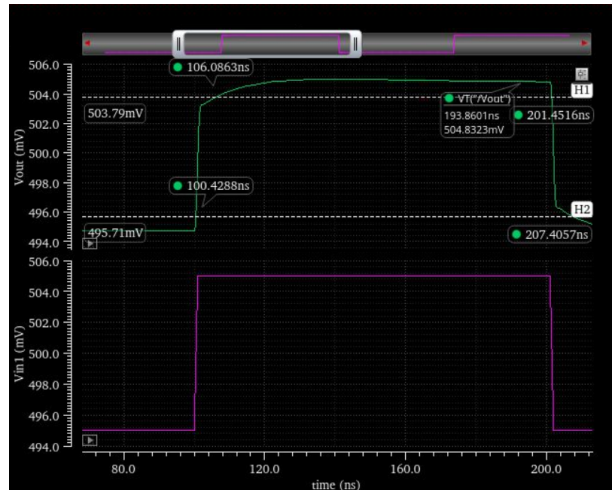


Figure 3.9: Annotated 10-90% times (495mV–505mV)

#### 3.5.2 0.505V - 0.495V

Calculating the 10 to 90% settling time as seen in figure 3.9:

$$106.8429\text{ns} - 100.4577\text{ns} = 6.3852\text{ns}$$

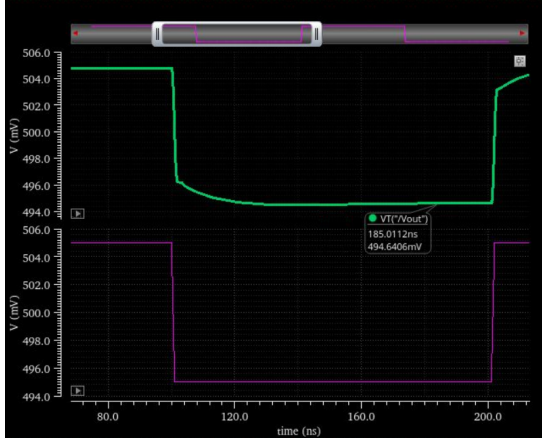


Figure 3.10: Closed-loop step response (505mV–495mV)



Figure 3.11: Annotated 10-90% times (505mV–495mV)

### 3.5.3 0V - 1V

Calculating the initial rising slope of the output as seen in figure 3.11:

$$SR = \frac{285.249\text{mV} - 208.8412\text{mV}}{101.2472\text{ns} - 100.8223\text{ns}} = 179.8 \text{ mV/ns} = 179.8 \text{ V/}\mu\text{s}$$

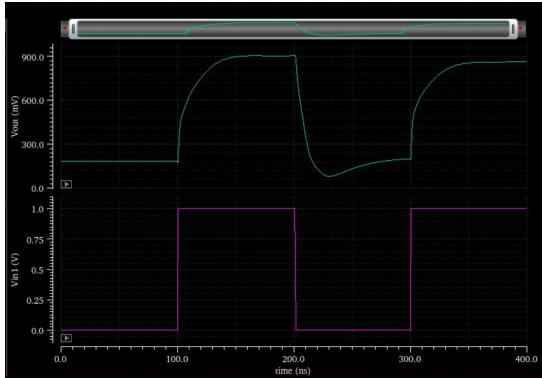


Figure 3.12: Closed-loop step response (0–1 V)

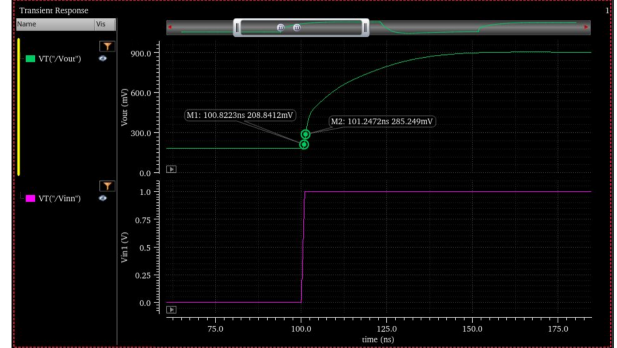


Figure 3.13: Annotated initial rising slope (0–1 V)

### 3.5.4 1V - 0V

Calculating the initial falling slope of the output as seen in figure 3.13:

$$SR = \frac{759.0422\text{mV} - 814.2533\text{mV}}{102.0432\text{ns} - 101.4839\text{ns}} = -98.715 \text{ mV/ns} = -98.715 \text{ V/}\mu\text{s}$$

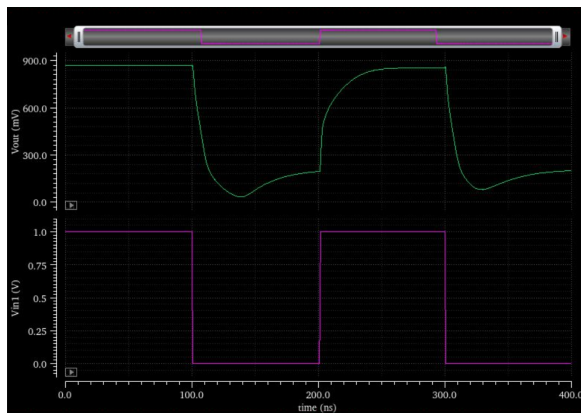


Figure 3.14: Closed-loop step response (1-0 V)

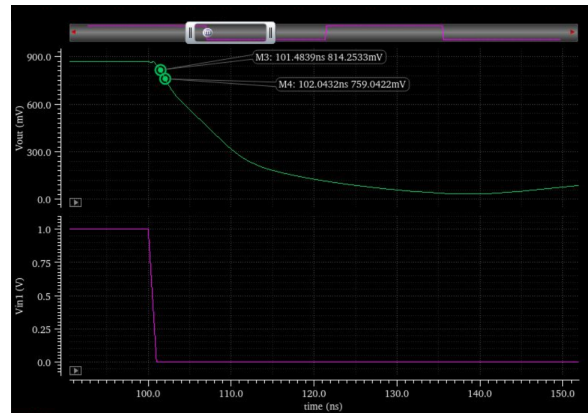


Figure 3.15: Annotated initial falling slope (1-0 V)

# Chapter 4

## Conclusion

The two design requirements not met were namely the nominal output CM voltage and the differential output swing. The nominal output CM voltage fell short by 0.05V at 446mV rather than 500mV. Meanwhile, the differential output swing fell short by 50mV at 0.65V compared to the requirement of  $\geq 0.75V$ . In order to improve on the design, we can increase the upper bound of the output voltage limit by increasing the gate voltage of the second stage pmos device. This would require increasing the drain-source voltage of the second stage nmos by decreasing W/L to drop the current through stage 2. As a result, transconductance will drop, but output resistance will increase, hence the loss in gain within stage 2 should be minimal.

Overall, This project provided valuable insights into the practical challenges of designing analog CMOS circuits. Through designing with EDA tools such as Cadence, it was highly advantageous to run 100 parameter simulations to examine the effects of changing parameters and optimising the design. The experience gained through this design process will be directly applicable to more advanced analog and mixed-signal IC designs.